Quant Problem Solutions

A working notebook of problems and solutions

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Preface

This document is a personal collection of quantitative problems I have worked through, along with my solutions. The goals of this notebook are:

- To force myself to write up clean, end-to-end solutions rather than scratch work.
- To tag and categorize problems so I can revisit weak areas.

Each problem is presented with a difficulty rating, tags, topics, the problem statement, and my full solution. Note that the problems are not in any particular order and are just in the order of when I solved them. If I'm for some reason not allowed to show certain problems here, please contact me at cosmoyangwu@gmail.com. Enjoy!

Contents

MEDIUM Tags: brainteasers Topics: Events Date: May 4, 2026

1. Repeated Flipper

Problem

25 fair coins are lined up and flipped once. Afterwards, remove all of the coins that showed tails, and repeat the aforementioned process until you have no coins left or all of the coins show up heads. Find the probability that you end the game by having all of the coins show heads.

Solution. Note that for the game to end, you must either have all heads or all tails at the final step. There is intrinsic symmetry, so our answer is simply $\boxed{\frac{1}{2}}$.

MEDIUM Tags: Brainteasers Topics: Other Date: May 4, 2026

2. Sum1

Problem

Find two real numbers x and y composed of only 1s in their decimal expansion such that $xy = x + y$. What is xy?

Solution. To solve, first note we can rewrite this as $x = \frac{y}{y-1}$. We must then use the property of x first to note that our expression must be an expression of 1s. We use this fact to know that $y = 1.1$ because otherwise, we wouldn't get all 1s for x. Thus, $x = 11$ and so $xy = 12.1$. The key is to isolate effects to utilize known properties.

MEDIUM Tags: Brainteasers Topics: Other Date: May 4, 2026

3. Mathematical Birthday

Problem

Your grandpa from the 19th and 20th centuries stated that he was $x^2 + y^2$ years old in the year $x^4 + y^4$, $2m$ years old in the year $2m^2$, and $3n$ years old in the year $3n^4$. All the variables are integers. Determine the value of $xymn$.

Solution. First, to get $x^4 + y^4$ between 1900 and 2000, we must have $x = 5$ and $y = 6$ (the other way around works too, since we have symmetric equations! :D). Then, we have $625 + 1296 = 1921$, and so our grandpa is 61 years old, meaning he was born in 1860. We then must have $1860 + 2m = 2m^2$, which gives us $m = 31$. Similarly, $1860 + 3n = 3n^4$, which yields $n = 5$. Thus, our answer is $5 \times 5 \times 6 \times 31 = \boxed{4650}$.

MEDIUM Tags: Brainteasers Topics: Combinatorics Date: May 4, 2026

4. Picture Day

Problem

Ten students of distinct heights are lining up for a picture. The photographer requires that the two tallest students stand in the two center positions and that the remaining students line up such that the heights strictly descend outwards. How many ways can the students line up?

Solution. This is a classic constructive counting problem! First, for our center students, we have 2 ways to line them up. Next, for each of the remaining 8 students, consider choosing 4 of them for the left side. Since the height must be in correct order, how the 4 students for each side are ordered is already determined. Thus, our answer is $2 \cdot \binom{8}{4} = \boxed{140}$.

MEDIUM Tags: Probability Topics: Combinaorics Date: May 4, 2026

5. Ace Distribution

Problem

Assume the dealer holds a fair deck of 52 cards. She shuffles the deck and gives each of you and your three friends at the table 13 cards. What is the probability that each of you receive an ace?

Solution. Suppose the first card for each of us is an ace. Then, for the remaining cards, we can split them up in $\frac{48!}{12!^4}$ ways. This is simply given from splitting up the 48 cards into each of the friends and then accounting for the fact that order doesn't matter within one person's cards. The total number of ways is then $\frac{52!}{13!^4}$. We have $4! = 24$ ways of how each of us gets an ace (like what suit and what not). Thus, our answer is $24 \cdot \frac{48! \cdot 13!^4}{52! \cdot 12!^4} = \boxed{\frac{2197}{20825}}$

MEDIUM Tags: Brainteasers Topics: Other Date: May 4, 2026

6. Say 50!

Problem

You and your friend are playing a game where you each pick an integer from 1 to 10 inclusive and add that to a running total. The person that states 50 is the winner. You get to go first, what number do you say first?

Solution. The key to solving this problem is to realize that we can start from the bottom up. If the game is to just say 10, then if you start first, you're guaranteed to win. If you're starting at 11, then you're guaranteed to lose. This includes all multiples of 11. Meaning, if you start at 44, you will lose. Thus, if we say $\boxed{6}$, my friend will be essentially starting at 44 and so will lose!

EASY Tags: Brainteasers Topics: Combinatorics Date: May 5, 2026

7. 2D Paths I

Problem

You are playing a 2D game where you are trapped within a 6×6 grid and navigate via points on the grid. You start at the bottom left: point $(0, 0)$. You can only move up and right. How many possible paths can you take to get to the top right point: $(6, 6)$?

Solution. First, note that for each path, there is a $1-1$ correspondence to how you arrange 6 letters of R and 6 letters of U. Thus, our answer is $\binom{12}{6} = \boxed{924}$.

EASY Tags: Probability Topics: Discrete Random Variables, Expected Value Date: May 5, 2026

8. Coin Runs

Problem

You toss a fair coin 100 times and record the outcomes. How many runs will you have on average? A run is classified as the longest continuous flips of heads or tails.

Solution. I sadly got this question wrong when solving it. However, the solution is to consider adjacent coins X_i and X_{i+1}. Between 2 coins, the probability that X_{i+1} is a new coin is $\frac{1}{2}$. Thus, for $X_2, X_3, ...X_{100}$, the probability that each is a new run is $\frac{1}{2}$, and for X_1, the probability is 1. Thus, by summation of expectation, the number of runs on average would be $1 + \frac{99}{2} = \boxed{50.5}$.

EASY Tags: Brainteasers Topics: Other Date: May 5, 2026

9. Revolver Time

Problem

Jenny is firing her revolver, which has 6 chambers. The time from the first shot to the last is 1 minute. Assuming a uniform firing rate, how many seconds will it take Jenny to fire 3 shots?

Solution. We have 5 gaps, meaning each turn takes $\frac{60}{5} = 12$ seconds, and so it would take $\boxed{24}$ seconds.

MEDIUM Tags: Brainteasers Topics: Games Date: May 6, 2026

10. Greedy Pirates

Problem

A treasure of 100 gold coins must be divided among five pirates, ranked by seniority from 1 (least senior) to 5 (most senior). They follow these rules to split the coins:

1. The most senior pirate proposes how many coins to distribute to each pirate.
2. All pirates vote on the proposal, including the pirate that made the proposal. If 50% or more of the pirates accept the proposal, the coins are distributed as proposed.
3. Otherwise, if more than 50% of the pirates reject the proposal, the most senior pirate is tossed overboard. The process restarts with the next most senior pirate making a proposal and the remaining pirates casting votes.
4. This process repeats until a proposal is accepted or only one pirate remains, in which case the last pirate gets all 100 gold coins.

Pirates prioritize survival over wealth. The pirates will act rationally to maximize their gain while ensuring they survive. How many coins does the most senior pirate (pirate 5) get if he makes an optimal proposal? If the most senior pirate is always thrown overboard, answer 0.

Solution. This is another classic "induction" question where we must consider the case of 1 pirate, then 2 pirates, and etc. For 1 and 2, the most senior pirate can always get all the coins. If we have 3 people, then we would want the most senior to give person 1 1 coin so that he doesn't throw person 3 (most senior) out. This is the case since if the most senior does get thrown out, person 1 will end up getting 0 (person 2 will take all of it). Similarly, if we have 4 people, the most senior would want to give a coin to person 2. With this logic, person 5 would give a coin to both 1 and 3. So our answer is $\boxed{98}$.

MEDIUM Tags: Statistics Topics: Events Date: May 6, 2026

11. German Tanks

Problem

Suppose that German tanks are assigned distinct serial numbers $1, 2, \dots N$. You observe 6 tanks with serial numbers $38, 17, 59, 42, 97, and 120$. Under a frequentist approach, what is the best guess for N?

Solution. I also was unable to solve this problem successfully :(. However, the key to this problem is to notice that we can reframe this problem as an ace problem where the 120 serves as one of the checkpoints. Since we chose 5 other numbers before 120, that means we have $114/6 = 19$ as our gap. Thus, our closest value is $N = 120 + 19 = \boxed{139}$.

MEDIUM Tags: Probability Topics: Combinatorics, Conditional Probability Date: May 6, 2026

12. Consecutive Tails

Problem

A coin is flipped 10 times and the outcomes are recorded. Find the probability that any tails occur only in consecutive pairs. For example, with 4 flips, $TTHH, TTTT$, and $HHHH$ are both valid, but $HTHH$ and $TTTH$ are not valid.

Solution. Let's think about this with casework and just divide by 2^{10} at the end: If 0 pairs of tails, then 1 for this to occur.
If we have 1 pair only, we have 9 spots for the pair.
If we have 2 pairs, we can think of it as 2 groups along with 6 heads that we need to arrange in a line. This gives us 8 choose 2, which is 28.
For 3 pairs, we have 3 groups and 4 heads, so $\binom{7}{3} = 35$. For 4 pairs, we have 4 groups and 2 heads, so 15 ways.
For 5, we have 5 groups, so 1 way.
Our answer is then $(10 + 36 + 28 + 1)/1024 = \boxed{\frac{89}{1024}}$

MEDIUM Tags: Brainteasers Topics: Other Date: May 6, 2026

13. Paired Pumpkins II

Problem

Dracula has 5 pumpkins. When he pairs any two pumpkins, they weigh $21, 22, \ldots, 30$ pounds. What's the sum of the weights of all the pumpkins?

Solution. Consider each of the pumpkins. Note that it can be paired with one of the other 4 pumpkins, meaning each pumpkin is used a total of 4 times for weighing. This means we simply need to add up the weights and divide by 4 to get $\boxed{\frac{255}{4}}$ pounds as our answer.

HARD Tags: Probability Topics: Continuous Random Variables Date: May 7, 2026

14. Square Ratio

Problem

Suppose that $X, Y \sim \text{Unif}(0, 1)$ i.i.d. Compute the probability that $\left\lceil \frac{X}{Y} \right\rceil$ is a perfect square. The answer is in the form $q - \frac{\pi^2}{a}$ for a rational q and an integer a. Find aq.

Solution. Note that we have a π^2 in the desired form, and that probably means that it's of interest to use geometric probability. Recall that $\lceil \frac{Y}{X} \rceil = k$ implies that $k - 1 < \frac{Y}{X} \leqslant k$. Since we want the region where $k = n^2$, our equation of interest turns into $(n^2 - 1)X < Y \leqslant n^2 X$. Now, n can be any value, and so we must evaluate the area created by all values of n. For a particular value of n, we have that area to be

$$\int_0^1 \left(\frac{1}{n^2 - 1} - \frac{1}{n^2}\right) Y \, dY = \frac{1}{2}\left(\frac{1}{2}\left(\frac{1}{n - 1} - \frac{1}{n + 1}\right) - \frac{1}{n^2}\right)$$

for $n \geqslant 2$ (we used partial decomposition of fractions at the end). As for $n = 1$, we know that the area is simply $\frac{1}{2}$.

Now, we sum our expression from 2 to infinity. We see that our telescoping series evaluates to $\frac{1}{4} \cdot \frac{3}{2}$ while our other series sums up to $\frac{1}{2}(\frac{\pi^2}{6} - 1)$. Our sum is thus,

$$\frac{1}{2} + \frac{3}{8} - (\frac{\pi^2}{12} - \frac{1}{2}) = \frac{11}{8} - \frac{\pi^2}{12} \to a = 12, q = \frac{11}{8}$$

$$aq = \boxed{\frac{33}{8}}$$

as desired.

HARD Tags: Probability Topics: Expected Value Date: May 8, 2026

15. Decreasing Uniform Chain

Problem

Let $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ i.i.d. Let N be the first index n where $X_n \neq \min\{X_1, \dots, X_n\}$. Find $\mathbb{E}[X_{N-1}]$, i.e., the smallest value among the first N values selected. The answer will be in the form $a + be$ for integers a and b. Note here that e is Euler's constant. Find $a + b$.

Solution. This problem was very difficult, and I don't think I actually live-solved it :(. But here's the approach AI and I came up with:

Suppose $N = n$. We must first find the probability density function of X_{n-1}, let this be $f_{n-1}(y)$. Suppose the value of X_{n-1} is y. Note that the probability that the past $n - 2$ X_i being greater than X_{n-1} is equal to $(1 - y)^{n-2}$, and for those terms to be in decreasing order, this would make our probability be $\frac{1}{(n-2)!}(1 - y)^{n-2}$ (think about the ordering of the $n - 2$ variables). Finally, we must consider X_n, which has a probability of $1 - y$ of being greater than X_{n-1}.

This gives our probability density function as $f_{n-1}(y) = 1 \cdot \frac{(1-y)^{n-1}}{(n-2)!}$ where the 1 comes from the original PDF of X_{n-1}.

Recall that n can be any integer $n \geqslant 2$. Thus, our actual probability density function is

$$f(y) = \sum_{2}^{\infty} \frac{(1 - y)^{n-1}}{(n - 2)!} = (1 - y)e^{1-y}$$

where the last step is noticing that the summation with one of the $(1 - y)$ factors is the taylor series for e^x.

We now compute the integral to find our answer, which is

$$\mathbb{E}[X_{N-1}] = \int_0^1 y(1 - y)e^{1-y} \, dy = 3 - e$$

from integration by parts. Thus, our answer is $3 - 1 = \boxed{2}$.

HARD Tags: Probability Topics: Expected Value Date: May 9, 2026

16. Collecting Toys II

Problem

Every box of cereal contains one toy from a group of 5 distinct toys, each of which is mutually independent from the others and is equally likely to be within a given box. How many distinct toys can you expect to collect if you bought 7 boxes?

Solution. The key to solving this problem is to assign an indicator for each of the toys $I_1, I_2, ..I_5$, where $I_i = 1$ if it does appear in the 7 boxes. We now simply find the expected value of all these indicators. For I_i, the probability that it does appear is $1 - (\frac{4}{5})^7$ via complementary probability. Thus, our answer is simply $\boxed{5(1 - (\frac{4}{5})^7)}$.

HARD Tags: Probability Topics: Expected Value Date: May 9/10, 2026

17. Close Dice II

Problem

On average, how many rolls of a fair 6-sided die are required to get two consecutive rolls with a difference of at most 1?

Solution. I didn't solve this all the way through because it got too tedious. Anyways, this is a classic Markov Chain problem. Suppose S_i is the expected number of rolls after rolling the number i. By symmetry, we have $S_1 = S_6$, $S_2 = S_5$, $S_3 = S_4$. And the following equations:

$$S_1 = 1 + \frac{1}{6}(S_3 + S_4 + S_5 + S_6) = 1 + \frac{1}{6}(2S_3 + S_2 + S_1)$$

$$S_2 = 1 + \frac{1}{6}(S_4 + S_5 + S_6) = 1 + \frac{1}{6}(S_1 + S_2 + S_3)$$

$$S_3 = 1 + \frac{1}{6}(S_1 + S_5 + S_6) = 1 + \frac{1}{6}(2S_1 + S_2)$$

and solving this system of equation yields $S_1 = \frac{288}{115}, S_2 = \frac{246}{115}, S_3 = \frac{252}{115}$. Thus, our answer is $E = 1 + \frac{1}{6}(S_1 + S_2 + S_3 + S_4 + S_5 + S_6) = 1 + \frac{1}{3}(S_1 + S_2 + S_3) = \boxed{\frac{377}{115}}$.

MEDIUM Tags: Probability Topics: Conditional probability, discrete random variable Date: May 11, 2026

18. Odd Coin Flips

Problem

You flip 100 fair coins simultaneously. What is the probability that you observe an odd number of heads?

Solution. Let's let our probability for getting odd heads for n coin flips be $p(n)$, then let's condition it at our first coin flip, if we get heads, then we have $\frac{1}{2}(1 - p(n-1))$ of getting odd number of heads in total since we want the next $n-1$ flips to give even number of heads. If we get tails, then we have $\frac{1}{2}p(n-1)$ since we just want the next $n-1$ flips to have an odd number of heads. Adding these up, we get $p(n) = 1/2$ always.

MEDIUM Tags: Probability Topics: Combinatorics, Events Date: May 11, 2026

19. Paired Values I

Problem

Suppose we have the values $1-6$ in a bowl. We draw them without replacement, noting the order in which we selected them. We multiply the first two values together, the next two values together, and the last two values together. Lastly, we add the three products above. Find the probability the sum is odd.

Solution. First, note that there's a total of $6! = 720$ ways of different ways to draw the values out. We consider constructive counting and consider how we can get an odd number by considering the 3 groups as group 1, group 2, and group 3. Note that to get an odd sum, there must be one group with 2 even numbers; otherwise, we are simply adding 3 even numbers together! Suppose group 1 has the 2 even numbers, group 2 has 1 even number, and group 3 has 0. Group 1 has 3×2 ways of putting 2 even numbers, while group 2 has 3×2 ways of choosing 1 of the 3 odd numbers and arranging it with the other even number. Finally, we have 2 ways of arranging the last two odd numbers in group 3. We can also arrange the 3 groups in $3! = 6$ ways. Thus, our answer is $\frac{6\times6\times2\times6}{720} = \boxed{\frac{3}{5}}$

EASY Tags: Probability Topics: Events Date: May 12, 2026

20. Poor Odds

Problem

Angelina is playing a game which she wins with probability 0.1. She must pay 10 to play, and if she wins, she receives 80. If Angelina starts out with 30, to the nearest thousandth, what is the probability that she wins exactly once before losing it all?

Solution. This is a typcial negative binomial random variable. We must have $WLLLLLLLLLL$ where W can take any of the first 3 positions. This gives us $3 \times 0.1 \times 0.9^10 = 0.0729 \times 4 = 0.105$

EASY Tags: Pure Math Topics: Linear Algebra Date: May 12, 2026

21. Orthogonal Cosine

Problem

Let x and y be vectors in $\mathbb{R}^n$ with angle 75° between them. Let A be an orthogonal $n \times n$ matrix. Find the angle between Ax and Ay.

Solution. Note that $Ax \cdot Ay = (Ax)^T Ay = x^T A^T Ay = x^T y = x \cdot y$ where we used the fact that $A^T A = I$ since it is an orthogonal matrix. Thus, the angle is also just 75.

EASY Tags: Pure Math Topics: Stochastic Processes Date: May 12, 2026

22. Make Your Martingale I

Problem

Let $X_1, X_2, \ldots$ be IID random variables that take the values ± 1 with equal probability. Fix a constant c. Let $M_0 = 1$ and $M_n = M_{n-1} e^{X_n + c}$. There is a unique value of c such that M_n is a martingale with respect to the natural filtration $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Find an expression for c, the answer will be in the form $-\log\left(ae^b + ce^d\right)$ for values a, b, c, d. Note that log here refers to the natural logarithm. Find $a + b + c + d$.

Solution. By definition of a martingale, we have

$$\mathbb{E}(M_{n+1} \mid \mathcal{F}_n) = M_n$$

which using our information gives us

$$M_n \mathbb{E}(e^{X_{n+1}+c} \mid \mathcal{F}_n) = M_n$$

and since X_n is an independent probability distribution and $M_n \neq 0$ (because $M_0 = 1$), we have

$$e^c \mathbb{E}(e^{X_n}) = 1$$

but we know

$$\mathbb{E}(e^{X_N}) = \frac{1}{2}(e^{-1} + e^1)$$

and so substituing this back in, we get

$$c = -\log\left(\frac{1}{2}(e + e^{-1})\right)$$

and so our answer is $1 - 1 + 2 \cdot 1/2 = \boxed{1}$

HARD Tags: Probability Topics: Expected Value Date: May 13, 2026

23. Shattering Orbs

Problem

7 orbs are labeled $1-7$ and are linked linearly in a vertical stack from the ceiling with orb 1 being a part of the ceiling and orb 7 being closest to the floor. Each orb is attached to adjacent orbs by a chain link. At each time step, one of the remaining links is going to be uniformly at random selected and cut. As a result, all the orbs below that link will fall and shatter. What is the expected number of cuts needed until orb 1 is the only remaining orb?

Solution. This is actually a very simple problem. The key is to notice we can do this recursively with the formula

$$E(n) = 1 + \frac{1}{n}(E(1) + E(2) + ... + E(n-1)), E(0) = 0$$

we comptue this recursively to get $E(6) = \frac{49}{20}$ as desired.

EASY Tags: Finance Topics: Finance Date: May 13, 2026

24. Company Purchase I

Problem

How much would you pay for a company that generates \$100 of cash flow every single year, until you sell, if you have a targeted yield of 16%? Enter your solution rounded to the nearest whole number.

Solution. This is just a classical present value; we have $0.16x = 100$ and then solve for $x = 625$.

EASY Tags: Probability Topics: Events, Continuous Random Variables Date: May 13, 2026

25. Points on a Circle III

Problem

There are 8 points on a circle. You draw all possible chords that can be formed by connecting each pair of points. Then, you randomly selects 4 chords. What is the probability that the 4 chords form a convex quadrilateral?

Solution. Note that we have $\binom{8}{2} = 28$ chords and so we have $\binom{28}{4}$ ways of choosing 4 chords. Out of those, only $\binom{8}{4}$ of those would become a convex quadrilateral. We see this since any 4 points create a convex quad, and we have 8 points. Thus, our answer is just $\binom{8}{4}/\binom{28}{4} = \boxed{\frac{2}{585}}$ by considering the number of successful outcomes divided by the total number of outcomes.

EASY Tags: Probability Topics: Conditional Probability Date: May 13, 2026

26. Cheater

Problem

As a good test taker, on a multiple choice exam with 5 options per question, Gabe either knows the answer to a question beforehand or chooses an answer completely at random. If he knows the answer beforehand, he selects the correct answer. If the probability that Gabe knows the answer to any given question is 0.6, find the probability that an answer that was correct was one for which he knew the answer.

Solution. This is just a typical conditional probability question. We have

$$P(\text{correct}) = P(\text{knows})P(\text{correct}|\text{knows}) + P(\text{don't know})P(\text{correct}|\text{don't know})$$

$$P(\text{correct}) = (0.6)(1) + (0.4)(0.2) = 0.68$$

Thus, our answer is $P(\text{correct and knows})/P(\text{correct}) = \boxed{15/17}$

HARD Tags: Pure Math Topics: Linear Algebra Date: May 15, 2026

27. Non-Zero Eigenvalue

Problem

Let $\boldsymbol{x}_n = \begin{bmatrix} 1 & 2 & \dots & n \end{bmatrix}^T \in \mathbb{R}^n$. Define $\boldsymbol{A}_n = \boldsymbol{x}_n \boldsymbol{x}_n^T$. For any fixed n, there is a single non-zero eigenvalue. Call this eigenvalue λ_n. Find the value of k such that $\dfrac{\lambda_n}{n^k}$ converges to a finite non-zero value.

Solution. We use the intrinsic nature of A_n, note that $A_n^2 = x_n(x_n^T x_n)x_n^T = (1^2 + 2^2 + ... + n^2)A_n = (n)(n+1)(2n+1)/6A_n$, this means that we have

$$A_n^2 \bar{v} = \lambda_n^2 \bar{v}$$

but using our derivation, we note

$$\frac{n(n+1)(2n+1)}{6} A_n \bar{v} = \lambda_n^2 \bar{v} = \frac{n(n+1)(2n+1)}{6} \lambda_n \bar{v}$$

Thus, the nonzero eigenvalue is of polynomial degree 3, and our answer is $k = 3$

HARD Tags: Probability Topics: Expected Value, Continuous Random Variables Date: May 16, 2026

28. Circular Slice I

Problem

A random angle $\theta_1 \sim \text{Unif}(0, 2\pi)$ is selected. The arc of the unit circle that sweeps out θ_1 radians is marked red going counterclockwise starting from $(1, 0)$. Two other angles $\theta_2, \alpha \sim \text{Unif}(0, 2\pi)$ IID (independently, identically distributed) are also selected.

Afterwards, an arc of length θ_2 radians starting from the point that is α radians counterclockwise of $(1, 0)$ is swept out and colored blue. What is the probability that the blue and red regions are disjoint?

Solution. The key is to first scale everything from 2π to 1 for the circumference to make the calculations easier. Now, consider fixed values of $u = \theta_1$ and $v = \theta_2$, then $w = \alpha$ needs to be $w > u$ and $w + v < 1$ to avoid a wrap-around. This means out of an interval of length 1, w must be in the interval $1 - u - v$. Otherwise, the condition cannot be satisfied. At the same time, note that $v \leqslant 1 - u$ is required, or else no w works. Our probabiltiy is then

$$\int_0^1 \int_0^{1-u} (1 - u - v)\ dv\ du = \boxed{\frac{1}{6}}$$

as desired.

HARD **Tags: Brainteasers** **Topics: Pure Math** **Date: May 17, 2026**

29. Shuffled Deck

Problem

Suppose we have a standard deck of cards arranged in some order. We shuffle the cards by cutting the deck into two halves: cards in positions 1–26 and 27–52. Then, we alternate cards from each half of the deck. Namely, the cards in positions 1–26 are now in positions $2, 4, \ldots, 52$, and the cards in positions 27–52 are now in positions $1, 3, \ldots, 51$. What is the minimum number of shuffles needed before the deck returns to its original state?

Solution. The key to this problem is to note that after each shuffle, the card initially at position x is now at position $2x \equiv y \pmod{53}$. This means we want n such that

$$2^n x \equiv y \pmod{53}$$

which is equivalent to finding the smallest n such that

$$2^n \equiv 1 \pmod{53}$$

By Fermat's little theorem, this has to be a factor of $53 - 1 = 52$. (**I need to review why this is the case**). Checking the values, we would see that the smallest number that works for n is $\boxed{52}$, so that is our answer.

HARD **Tags: Probability** **Topics: Continuous Random Variable, Expected Value** **Date: May 18, 2026**

30. Circular Cut

Problem

We select 3 points uniformly and randomly from the circumference of the unit circle. These 3 points divide the circle into 3 arcs. Find the expected length of the arc that contains the point $(1, 0)$. The answer is in the form $q\pi$ for a rational number q. Find q.

Solution. Let's first consider the arc at point $(1,0)$. It will have a counterclockwise and clockwise component, which are equal in expectation by symmetry. So let's consider the counterclockwise piece first. We want

$$\mathbb{E}(L) = \int_0^{2\pi} \mathbb{P}(L \geqslant x)\ dx$$

where L is the length of the counterclockwise component. The probability is simply the probability that all 3 points are not on the arc of length x, ie $(\frac{2\pi - x}{2\pi})^3$. Then, we integrate to get

$$\mathbb{E}(L) = \int_0^{2\pi} (\frac{2\pi - x}{2\pi})^3\ dx = \frac{\pi}{2}$$

Since we have 2 components, counterclockwise and clockwise, our answer is $2 \cdot \pi/2 = \boxed{\pi}$, and so the fraction component is just 1.

MEDIUM Tags: Brainteaser Topics: None Date: May 19, 2026

31. Cat Dog Line

Problem

You're walking down an infinite street, and you notice that six out of every seven cats on a sidewalk are followed by a dog, while one out of every four dogs is followed by a cat. What proportion of animals on the sidewalk are dogs?

Solution. Let there be a total of c cats and d total dogs. Consider the transitions from cats to dogs and dogs to cats. Note that since we are on an infinite street, the number of transitions for those two must be equal. Then, we have

$$\frac{6}{7}c = \frac{1}{4}d \rightarrow c = \frac{7}{24}d$$

and so we have

$$d/(c+d) = \boxed{24/31}$$

as our answer.

MEDIUM Tags: Statistics Topics: Linear Regression Date: May 19, 2026

32. Coefficient Swap

Problem

Suppose that we have two datasets X and Y with $\mathrm{Var}(X) = 10$ and $\mathrm{Var}(Y) = 20$. We perform the linear regression $y \sim \alpha_x + \beta_x x$ and obtain $\beta_x = 1$. Suppose now that we perform the regression $x \sim \alpha_y + \beta_y y$. Find β_y. If the value can't be determined, enter -100.

Solution. Recall that $\beta_x = r\frac{s_y}{s_x}$ and so this means $r = 1/\sqrt{2}$ since $s_y = \sqrt{10}$ and $s_x = \sqrt{20}$. Then, $\beta_y = r\frac{s_x}{s_y} = \boxed{1/2}$

EASY Tags: Probability Topics: Conditional Probability Date: May 20, 2026

33. Unknown Baby

Problem

There are 5 babies in a room: 2 boys and 3 girls. one baby with unknown sex is added. One baby is random selected and it is a boy. What's the probability that the added baby is a boy? You may assume that each gender is equally likely at birth.

Solution. This is a simple probability question. Note that the probability that we got a boy is

$$P = (1/2)(2/6) + (1/2)(3/6) = 5/12$$

and the probability of this happening and the baby added is a boy is $P = 3/12$. Thus, our answer is $3/5$

EASY Tags: Probability Topics: Discrete Random Variables Date: May 20, 2026

34. Minimal Flipping

Problem

3 people flip a weighted coin with probability of heads $p = 0.25$ until they obtain their first tails. Find the probability that none of the three people flip more than twice.

Solution. We can do this problem via complementary probability and note that the probability that all need more than twice is 0.25^2 for each person, meaning the answer is $(1 - 0.25^2)^3$ which is equal to 0.824

EASY Tags: Probability Topics: Expected Value Date: May 20, 2026

35. Ship Stops

Problem

You are sailing from one island to another. Uninterrupted travel between the two islands takes 1212 minutes. However, there are 55 wave checks between the islands. Four of the wave checks will stop you with 25% probability each. These wave checks will add 11 minute to your total travel time if you are stopped. However, the last wave check stops you with 75% probability. This wave check will add 44 minutes to your travel time if you are stopped. Find the expected duration (in minutes) of the trip between the two islands.

Solution. This is just $0.25 \cdot 4 + 0.75 \cdot 4 + 12 = 16$ via the basic expected value formula.

HARD Tags: Brainteasers Topics: None Date: May 21, 2026

36. Heaven 37

Problem

Let x be a 5 digit integer in the form $37abc$ such that $37abc, 37bca$ and $37cab$ are all divisible by 37. How many possible values of x are there?

Solution. This condition requires the following to all be $0 \mod 37$ Thus, we have

$$100a + 10b + c \equiv 0 \pmod{37}$$

$$100b + 10c + a \equiv 0 \pmod{37}$$

$$100c + 10a + b \equiv 0 \pmod{37}$$

But note that since $1000 \equiv 1 \pmod{37}$, if we multiply the first equation by 10 or 100, we can get the other 2 equivalences. Thus, fulfilling the first equation implies satisfying the other two equations. Then, we simply need to find all numbers that satisfy abc being a multiple of 37, but this is equivalent to asking how many numbers between 0 and 1000 are a multiple of 37. This is equal to $1 + 27$ from $0 \cdot 37$ to $27 \cdot 37 = 999$. Thus, our answer is $\boxed{28}$.

HARD Tags: Probability Topics: Expected Value Date: May 22, 2026

37. Sum Exceedance III

Problem

Let

$$X_1, X_2, \cdots \sim \text{Unif}(0, 1)$$

be IID random variables. Define

$$N_1 = \min\{n : X_1 + \cdots + X_n > 1\},$$

and

$$S_n = X_1 + \cdots + X_n.$$

Compute $\mathbb{E}[S_{N_1}]$.
The answer will be in the form ce for a rational number c. Find c.

Solution. The key is to note that N_1 is a stopping time. Thus, by Wald's identity, we have

$$\mathbb{E}(S_{N_1}) = \mathbb{E}(X)\mathbb{E}(N) = \frac{1}{2}\mathbb{E}(N)$$

Thus, our problem simplifies to finding

$$\mathbb{E}(N) = \sum_{n=0}^{\infty} \mathbb{P}(N > n) = \sum_{n=0}^{\infty} \mathbb{P}(S_n \leqslant 1)$$

Note that the probability for

$$X_1 + X_2 + ... + X_n \leqslant 1$$

is equivalent to the volume under an n-simplex, which is equal to $\frac{1}{n!}$ by induction. **This proof is worth revisiting.** Thus,

$$\mathbb{E}(N) = \sum_{n=0}^{\infty} \frac{1}{n!} = e$$

Therefore, $c = \boxed{\frac{1}{2}}$.

HARD Tags: Probability Topics: Combinatorics Date: May 23, 2026

38. Leftwards Frog

Problem

A frog is hopping between 8 lily pads labeled $1-8$ from left to right. The frog starts on lily pad 1. At each turn, the frog hops to a lily pad that it has not been to yet. Find the probability the frog made exactly 1 leftwards hop after visiting all of the stones.

Solution. **This problem involves a type of number called the Eulerian numbers. It is worth revisiting what those are.** The key to this problem is realizing that when there's only 1 leftward jump, we essentially have a partition of two subsets, and then reordering each subset into a strictly increasing tuple. There are 2^7 such partitions (each element can either go into subset A or subset B). We can imagine the frog jumping through the elements in A and then jumping through the elements in B. For example, if $A = \{2, 4, 7\}$ and $B = \{3, 5, 6, 8\}$, we would do our jumping in that particular order. However, we must ignore the partitions where the subsets have only consecutive integers like $A = \{2, 3, 4\}$ and $B = \{5, 6, 7, 8\}$, as this leads to no leftward jumps. There are 8 of those, so we must subtract those to get 120 valid arrangements. Thus, our answer is $120/7! = \boxed{\frac{1}{42}}$.

HARD Tags: Probability Topics: Combinatorics Date: May 24, 2026

39. Specific Partition

Problem

Let $S = \{1, 2, 3, \ldots, 22\}$. Find the number of ways in which S can be partitioned into eleven subsets such that each subset contains exactly two elements of S and the absolute difference between the two elements of a subset is 1 or 11.

Solution. The key is to notice that this problem depends on edge $11 - 12$ and whether we use it or not. If we use this edge, we only have 1 possible way of connecting the rest of the numbers. If we don't, we get a recurrence relationship $M(n) = M(n-1) + M(n-2)$

(think of the graph as a 2×11 ladder). We can think of $M(11)$ as our desired answer, for a length of 11 ladder. Note that if we remove the edge $11 - 22$ together, then we get $M(10)$, and if we remove $10 - 11$, we must pair $21 - 22$ together, so we have $M(9)$, hence the recurrence relationship. We have $M(1) = 1$ and $M(2) = 2$. Solving this, we get $144 + 1 = \boxed{145}$ as our answer.

HARD Tags: Probability Topics: Combinatorics Date: May 25, 2026

40. Non-Disjoint Subsets

Problem

Let A and B be uniformly at random selected subsets of $\{1, 2, 3, 4, 5\}$. Find the probability that A and B are not disjoint.

Solution. Let's compute the probability they're disjoint and subtract it from 1. We can case work through how many elements A has and then subset B from that.
$|A| = 0$: 32 ways for B
$|A| = 1$: 5 choices for A, 16 for B, and then we have $(10, 8)$ for 2, $(10, 4)$, $(5, 2)$, $(1, 1)$ for $3, 4, 5$. Thus, the number of ways is $32 + 80 + 80 + 40 + 10 + 1 = 243$ ways and $2^{10} = 1024$ possible subsets, so we have our answer as $1 - 243/1024 = \frac{781}{1024}$

HARD Tags: Probability Topics: Conditional Probability, Expectation, Discrete Random Variable Date: May 26, 2026

41. Non-Consecutive Sequence

Problem

Suppose that you flip a coin n times and it turns out that no consecutive heads appear in the sequence. Let $E(n)$ be the expected number of heads that appear given this information. Compute $\lim_{n\to\infty} \frac{E(n)}{n}$. The answer is in the form $\frac{a}{b+\sqrt{c}}$ for integers a, b, and c with c minimal. Find abc.

Solution. Suppose a_n is the number of ways of getting no consecutive coins. We can see we have the relationship of $a_n = a_{n-1} + a_{n-2}$ (think about what the nth coin is). We have $a_n = F_{n+2}$. Now suppose $H(n)$ represent the total number of heads summed over all sequences. We would then have $E(n) = \frac{H(n)}{a_n}$. Now, for all valid sequences, note that if we start with a T, then the number of heads for $H(n)$ is $H(n-1)$, and if we start with H, our number of heads from the first head is a_{n-2} and the number of heads from the rest of the sequence contributes $H(n-2)$. Thus, we have the relationship

$$H(n) = H(n-1) + a_{n-2} + H(n-2)$$

, now if we divide by a_n, note we'll get

$$E(n) = \frac{H(n)}{a_n} = \frac{a_{n-1}}{a_n}E(n-1) + \frac{a_{n-2}}{a_n} + \frac{a_{n-2}}{a_n}E(n-2)$$

Since $a_n = F_{n+2}$, the ratios of consecutive terms converge to the golden ratio. As $n \to \infty$,

$$\frac{a_{n-1}}{a_n} \to \frac{1}{\phi}, \qquad \frac{a_{n-2}}{a_n} \to \frac{1}{\phi^2},$$

where $\phi = \frac{1+\sqrt{5}}{2}$ satisfies $\phi^2 = \phi + 1$, equivalently $\frac{1}{\phi} + \frac{1}{\phi^2} = 1$.

Now suppose $E(n) \sim cn$, so for large n we have $E(n-1) \sim c(n-1)$ and $E(n-2) \sim c(n-2)$. Substituting into the recurrence:

$$cn = \frac{1}{\phi}c(n-1) + \frac{1}{\phi^2}c(n-2) + \frac{1}{\phi^2}.$$

Expand the right-hand side:

$$cn = cn\left(\frac{1}{\phi} + \frac{1}{\phi^2}\right) - c\left(\frac{1}{\phi} + \frac{2}{\phi^2}\right) + \frac{1}{\phi^2}.$$

Using $\frac{1}{\phi} + \frac{1}{\phi^2} = 1$, the leading cn terms cancel from both sides, leaving

$$0 = -c\left(\frac{1}{\phi} + \frac{2}{\phi^2}\right) + \frac{1}{\phi^2}.$$

Solving for c and multiplying through by ϕ^2:

$$c\,(\phi + 2) = 1 \quad \Longrightarrow \quad c = \frac{1}{\phi + 2}.$$

Finally, substitute $\phi = \frac{1+\sqrt{5}}{2}$:

$$\phi + 2 = \frac{1+\sqrt{5}}{2} + 2 = \frac{5+\sqrt{5}}{2},$$

$$c = \frac{2}{5+\sqrt{5}}.$$

Therefore

$$\lim_{n\to\infty} \frac{E(n)}{n} = \frac{2}{5+\sqrt{5}}.$$

With $a = 2$, $b = 5$, $c = 5$, we get $abc = \boxed{50}$.

HARD Tags: Probability Topics: Combinatorics, Expected Value Date: May 27, 2026

42. Maximize Head Ratio II

Problem

Say that you are flipping a fair coin where you can stop flipping whenever you want. Your goal is to maximize the ratio between the number of heads you get with the total number of flips. The ratios are in the form $a : 1$. What is the expectation of a when you follow the strategy of stopping when you have more heads than tails? Round your answer to the nearest thousandths.

Solution. Note that we always stop when we go from a tie to 1 more head than tails. Thus, the ratio is a is equal to

$$a = \frac{k+1}{2k+1}$$

where we were at k heads and k tails right before this. Now, we need to find the probability we stop at $2k+1$ and sum over all values of k. An important key to note is that the number of ways you can reach $2k+1$ is equal to

$$\binom{2k}{k} - \binom{2k}{k+1} = \frac{1}{k+1}\binom{2k}{k}$$

where $\binom{2k}{k+1}$ is the number of non-working combinations. Suppose we flipped $2k$ coins, and we reached 1 in heads $-$ tails. Then, if we flip the rest of the trajectory (heads to tails and tails to heads), we get a total of 2 more heads than tails. Note that for all trajectories of flipping $2k$ coins that result in an instance of 1 more heads than tails, there is a $1-1$ correspondence to a trajectory of reaching a total of 2 more heads than tails. Thus, that's how we obtain $\binom{2k}{k+1}$.

Then, the probability of this occurring is

$$\frac{1}{k+1}\binom{2k}{k}(\frac{1}{2})^{2k+1}$$

And so our desired result is

$$E(a) = \frac{1}{2}\sum_{k=0}^{\infty}\frac{1}{4^k(2k+1)}\binom{2k}{k} = \frac{1}{2}\arcsin(1) = \frac{\pi}{4}$$

as our answer.

HARD Tags: Pure Math Topics: Stochastic Calculus, Calculus Date: May 28, 2026

43. Positive Brownnian II

Problem

Let B_t be a standard Brownian Motion. Find $P[B_2 > 0, B_8 > 0]$.

Solution. First, note that

$$B_8 = B_2 + (B_8 - B_2)$$

and we know that $B_8 - B_2$ is independent of B_2 and vice versa. Let X be B_2 and Y be $B_8 - B_2$. We want $\mathbb{P}(X > 0, X + Y > 0)$. Since X and Y are normal distributions, we get rewrite them as $X = \sqrt{2}Z_1$ and $Y = \sqrt{6}Z_2$, and the area we're integrating over would be

$$R = \{Z_1 > 0\} \cap \{Z_1 + \sqrt{3}Z_2 > 0\}$$

and our area would be

$$P = \int\int_R \frac{1}{2\pi}e^{-(z_1^2+z_2^2)/2}\,dz_1\,dz_2$$

You then use polar coordinates and some stuff about the angular wedge to get the answer is $\frac{1}{3}$. I'm too lazy to type it up here.

HARD Tags: Brainteasers Topics: None Date: May 29, 2026

44. Terminating Sum

Problem

Evaluate $\sum_{k\in S}\frac{1}{k^3}$, where S is the set of all positive integers such that $\frac{1}{k}$ has a terminating decimal expansion. For example, $2\in S$, as $\frac{1}{2}=0.5$ has a finite expansion. However, $7\notin S$, as $\frac{1}{7}=0.\overline{142857}$ does not terminate.

Solution. The only k that works are combinations of 2 and 5. Thus, we have

$$(1+1/2^3+1/4^3+...)(1+1/5^3+1/5^6+...)=\frac{1}{7/8}\frac{1}{124/125}=\frac{250}{217}$$

as our answer. We multiply those 2 expressions to get every single possible combination.

HARD Tags: Brainteasers Topics: None Date: May 30, 2026

45. Prime Janitors

Problem

Janitors at a large trading company come to work to see 100 open doors. These doors are labeled 1 to 100. Throughout the day, janitors open/close doors based off of their number. The first janitor closes every door that's a multiple of 2. Then the next janitor closes every door that's a multiple of 3 unless it's currently closed, in which case he opens it back up. This will continue by having the kth janitor open/close all door numbers that are a multiple of the kth positive prime integer. How many doors are open at the end of the day?

Solution. The only doors that are still open are those with an even number of prime numbers. They can either have 0 distinct prime number (1) or 2 distinct since 4 distinct would have a greater value than 100 $(2\cdot3\cdot5\cdot7\geqslant100)$. We can do complementary counting. For 1 distinct prime, we have 25 from primes, 4 from prime squares, 2 from prime cubes, 4 from higher powers, so a total of 35. For 3, we have to count them up and get 8, so the answer is $100-8-35=57$.

MEDIUM Tags: Statistics Topics: None Date: May 31, 2026

46. Averaging Squares

Problem

Let $A=[1,2,\ldots,2023,1^2,2^2,\ldots,2023^2]$. What is the median of A?

Solution. This is an AMC problem lol. We want to find the 2023rd and 2024th element. When we're at 1936, we have passed 44 perfect squares, so we would be at the 1980th spot for 1936. Add 43 and 44 and divide by 2, we get $1936 + 43.5 = 1,979.5$ as our answer.

MEDIUM Tags: Probability Topics: Combinatorics Date: May 31, 2026

47. Odd Stars

Problem

How many non-negative odd integer solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 96?$$

The answer is in the form $\binom{n}{k}$ for $k < \frac{n}{2}$. Find nk.

Solution. Let x_i be $2y_i - 1$ where each y_i is a positive integer. Substituting and simplifying the equation, we get

$$y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 51$$

and now we have a simple stars and bars equation. We have 50 gaps and 5 separators, so the answer is $\binom{50}{5}$ and our answer is $\boxed{250}$.

MEDIUM Tags: Brainteasers Topics: Combinatorics Date: June 1, 2026

48. Adult Concert

Problem

$\frac{5}{12}$ of concert attendees are adults. A bus carrying 50 people arrives at the concert. Now, $\frac{11}{25}$ of concert attendees are adults. What is the minimum number of adults who could have been at the concert before the bus arrived?

Solution. Let there be originally x adults and T total people. Suppose we introduced y new adults afterward. We then have the equations

$$5T = 12x$$

$$11(T + 50) = 25(x + y)$$

Substituting the first equation into the second (eliminating T), we get

$$550 = 25y - 7/5x \rightarrow 550 \cdot 5 = 125 - 7x$$

For this equation to be true, $7x$ must also be a multiple of 125, which means x is a multiple of 125. Thus, our answer is 125.

MEDIUM Tags: Brainteasers Topics: Combinatorics Date: June 1, 2026

49. 9 Digit Sum

Problem

Find the sum of all 9-digit integers using all of the digits 1-9. The answer is in the form

$$k \cdot a! \cdot (10^b - 1)$$

for integers a, b, and k such that a and b are maximal. Find abk.

Solution. Let's consider the 1st digit to the 9th digit. For the first digit, if it is digit x, then it has 8! Ways of doing so. We have 9 different values, so it is $45 \times 8! \times 10^8$. Same for the other digits, so we can use geometric to get $5 \times 8! \times (10^9 - 1)$.

HARD Tags: Brainteasers Topics: Combinatorics Date: June 2, 2026

50. Square Shade

Problem

Consider a 2023×2023 grid. The grid squares from the middle row are shaded in (this is the 1012th row). What is the probability that a randomly selected rectangle contains at least one shaded square?

Solution. This is just the probability you choose a top from above the shaded row and one from below the shaded row, so $\frac{1012 \cdot 1011}{\binom{2023}{2}} = \boxed{\frac{1012}{2023}}$

MEDIUM Tags: Probability Topics: Games, Discrete Random Variable Date: June 3, 2026

51. High-Low

Problem

You roll a fair 100-sided die with values 1-100 on the sides and observe the value that appears. You now must guess whether the second roll will be at least as large as the first roll or not. If the second value that appears is y and you guess correctly, you receive \$$y$ payout. Otherwise, you receive nothing. Assuming optimal play, find the smallest integer t at which we guess that the second roll is lower than the first.

Solution. Suppose the value we get on the first dice is t, then if we guess that the second dice is greater than t, the probability is $(100 - t)/100$. We want to find the point of t where it is not worth betting that it will be greater, which gives us:

$$(100 + t)/2 \times (101 - t) \leqslant t(t - 1)/2$$

We then get

$$t \geqslant 72$$

so our answer is 72.

MEDIUM Tags: Probability Topics: Stochastic Processes, Conditional Expectation Date: June 3, 2026

52. Thorough Frog

Problem

A frog has 100 lily pads arranged in a circle. Each one has a distinct number $0, 1, \ldots, 99$ on it, arranged in increasing order counter-clockwise starting from 0. The frog starts at lily pad 0 and hops to the closest lily pad on the left or right of its current position with equal probability per hop. Find the probability that when the frog lands on lily pad 50 for the first time, it has visited every other lily pad not labeled 50.

Solution. Note that to get to 50, we must either reach 49 or 51 first. By symmetry, the probability of eaching 49 or 51 first are equal. WLOG, suppose we reach 49 first. The question is asking for the probability that we reach 51 before reaching 50. This is the same as the gambler's ruin problem, whether one end is 51 and the other is 50. We can remodel this as having endpoints at 0 and 99 and being positioned at 98. This has a probability of $\frac{1}{99}$ of going back to 0, so that is our answer.

Note that we got this by denoting p_i as the probability of going back to 0 if we started at index i. We have $p_0 = 1, p_{99} = 0$ and $p_i = \frac{1}{2}(p_{i-1} + p_{i+1})$.

MEDIUM Tags: Probability Topics: Expected Value, Continuous Random Variables, Calculus Date: June 4, 2026

53. Absolute Normal Difference

Problem

We have $X \sim N(0, 1)$ and $Y \sim N(0, 4)$ are independent. Compute $E[|Y - X|]$. The answer will be in the form

$$\left(\frac{K}{\pi}\right)^b$$

for rational b and K. Find bK.

Solution. Note that $Y - X \sim N(0, 5)$, and so we simply need to integrate the normal distribution from 0 to ∞, double it, and multiply it by $\sqrt{5}$ and we're done. We integrate the normal probability density from 0 to ∞ because of the absolute value sign, meaning each positive value in the probability density is now twice as dense. Thus, we desire

$$E(|X|) = \int_0^\infty 2x \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = \sqrt{\frac{2}{\pi}}$$

and so our answer is

$$(\frac{10}{\pi})^{\frac{1}{2}}$$

and thus the desired product is $\boxed{5}$.

MEDIUM Tags: Probability Topics: Combinatorics Date: June 4, 2026

54. Mean Babysitter

Problem

10 kids are really hungry! Their babysitter has 12 units of food to give. However, she decides she only wants to give 4 of the children food. How many ways can she distribute the food units such that 6 of the children are hungry (receive no food), and the other 4 children receive at least 1 unit of food each?

Solution. This is a constructive counting problem. We first choose 4 out of the 10 kids to give food to, and then we use stars and bars for the 12 items, which yields 11 choose 3, so our answer is

$$\binom{10}{4}\binom{11}{3} = 210 \times 165 = 34650$$

EASY Tags: Probability Topics: Expected Value Date: June 5, 2026

55. Side Add

Problem

Ab rolls a fair standard 6-sided die. Afterwards, given he rolls k on the first roll, Ab rolls a fair $(6+k)$-sided die for the second roll with the values $1, 2, \ldots, 6+k$. Find the expected value that appears on Ab's second roll.

Solution. In general, the expected value for a $6+k$ cube is $\frac{(7+k)}{2}$. We sum over all possible values of k and divide by 6. We have

$$\frac{1}{6}(21 + \frac{21}{2}) = \frac{21}{4}$$

EASY Tags: Probability Topics: Combinatorics Date: June 5, 2026

56. Colorful Cube

Problem

Consider coloring the 6 sides of a cube each a distinct color among red, blue, green, yellow, purple, and pink. Find the probability that the sides that are colored red, green, and blue all share a vertex of the cube (i.e. corner) in common.

Solution. Consider arbitrarily placing the red on one of the six sides. This always happens with probability 1. However, the probability that the blue face is next to it is $\frac{4}{5}$, and the probability that the green is next to those after it is $\frac{2}{4}$, so our answer is $\frac{2}{5}$. One can see this more easily by turning this cube into a flat and visualizing.

EASY Tags: Brainteasers Topics: Combinatorics Date: June 5, 2026

57. Zero Appearance

Problem

How many positive integers at most 10000 have the number 0 somewhere?

Solution. This is a complementary question; we have $9 + 9^2 + 9^3 + 9^4 = 9 + 81 + 729 + 6561 = 7,380$ that don't have a 0 anywhere, so our answer is $10000 - 7380 = \boxed{2,620}$

HARD Tags: Probability Topics: Combinatorics, Events Date: June 6, 2026

58. Random Subsets

Problem

Subsets A and B are chosen uniformly at random from the collection of all subsets of a set X of cardinality 5. What is the probability that A is a subset of B?

Solution. Suppose B has x elements, we have $\binom{5}{x}$ ways to do so, and we just need A to be one of the 2^x subsets. Thus, we have $\sum_{x=0}^{5} \binom{5}{x} 2^x = 1+10+10 \cdot 4+10 \cdot 8+5 \cdot 16+32 = 243$ ways of A being a subset of B. There are $2^5 \cdot 2^5$ total ways of choosing subsets for A and B, so our answer is $\frac{243}{1024}$.

MEDIUM Tags: Brainteasers Topics: None Date: June 7, 2026

59. Distinct Date I

Problem

Find the next date where all of the digits, when expressed in the form $MM/DD/YYYY$, are distinct. For example, 01/23/4567 would be a valid date. Express your answer in the form $MMDDYYYY$.

Solution. The year would be 2345
. Month would be 06
. Day would be 17
. We get this by considering the fact that the month and day must start with 0 or 1 and we want to use 2 in the year

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MEDIUM Tags: Probability Topics: Continuous Random Variable, Expected Value Date: June 7, 2026

60. Generational Wealth II

Problem

Suppose that you generate a uniformly random number in the interval $(0, 1)$. You can generate additional random numbers as many times as you want for a fee of \$0.05 per generation. The number of additional generations must be decided prior to generating your first number. Your payout is the maximum of all the numbers you generate. Assuming optimal strategy, what is your expected payout of this game?

Solution. Suppose we generate a total of t random numbers. We want our expected max to be greater than $0.05(t-1)$. We now simply consider the pdf of the max value and then compute its expected value and set it greater than $0.05(t-1)$. The pdf for the max value is $f(x) = \frac{d}{dx}(x^t) = tx^{t-1}$. Then, the expected value is equal to

$$\int_0^1 tx^t \, dx = \frac{t}{t+1}$$

Our expected payout as a function of t is

$$f(t) = \frac{t}{t+1} - 0.05(t-1)$$

We set the $f'(t) = \frac{1}{(t+1)^2} - 0.05$ to 0. We get $t = \sqrt{20} - 1$ which is between 3 and 4, so we compare $f(3)$ and $f(4)$, and we see that both are equal to 0.65 for $f(t)$, so that is our answer.

HARD Tags: Probability Topics: Conditional Probability Date: June 8, 2026

61. 20-30 Die Split III

Problem

Alice and Bob have fair 30-sided and 20sided dice, respectively. The goal for each player is to have the largest value on their die. Alice and Bob both roll their dice. However, Bob has the option to re-roll his die in the event that he is unhappy with the outcome. He can't see Alice's die beforehand. Bob then keeps the value of the new die roll. In the event of a tie, Bob is the winner. Assuming optimal play by Bob, find the probability Alice is the winner.

Solution. The optimal strategy for Bob to reroll is when it gets a value less than the expected value. The expected value is 10.5, so he should re-roll when the first roll is less than 11. Thus, we do case work: If the first row is greater than 11, then Bob doesn't reroll, so the probability is a summation based on what Bob gets $(11, 12, \ldots, 20)$, and so we have

$$\frac{1}{20}\frac{19}{30} + \cdots + \frac{1}{20}\frac{10}{30} = \frac{29}{120}$$

On the case where Bob rerolls, we sum up the total and then divide by 2. We have

$$\frac{1}{2}(\frac{1}{20}\frac{29}{30} + \ldots \frac{1}{20}\frac{10}{30}) = \frac{13}{40}$$

Adding those together yields $\frac{17}{30}$

HARD Tags: Probability Topics: Continuous Random Variable, Expected Value Date: June 9, 2026

62. Expected Chord Length

Problem

You uniformly select two random points from the circumference of the unit circle. Find the expected length of the chord (line segment) between the two points, given that the answer is in the form $x\pi$ for some rational number x. Find x.

Solution. The key to this problem is to first fix a point to $(1, 0)$. The other angle will go from 0 to 2π, but we only care about the 0 to π portion, so we can restrict the uniform distribution to become a uniform distribution from 0 to π. The length of the chord is equal to $2\sin(\theta/2)$, and so integrating this via

$$\frac{1}{\pi}\int_0^\pi 2\sin(\theta/2)\ d\theta$$

yields $4/\pi$, which is our answer.

HARD Tags: Probability Topics: Expected Value Date: June 10, 2026

63. Ramen Bowl

Problem

There are 100 noodles in your bowl of ramen. You take the ends of two noodles uniformly at random and connect the two, putting the connected noodle back into the bowl and continuing until there are no ends left to connect. On average, how many circles will you create? Round to the nearest whole number.

Solution. Suppose you have n noodles and randomly choose one of the edges. The probability that you choose the same noodle again is $\frac{1}{2n-1}$ to form a circle. Otherwise, we have a total of $n-1$ noodles now. Thus, we have

$$E(n) = \frac{1}{2n-1}(E(n-1)+1) + \frac{2n}{2n-1}(E(n-1)) = E(n-1) + \frac{1}{2n-1}$$

with $E(1) = 1$. Thus, we sum them up to get $E(100) \approx 3$.

HARD Tags: Probability Topics: Combinatorics Date: June 11, 2026

64. Three-Way Tile

Problem

How many ways can you tile a 3×8 grid with 1×2 and 2×1 tiles?

Solution. Let $t(n)$ be the number of ways to fill a $3 \times n$ grid, while $s(n)$ is the number of ways to fill a $3 \times n$ grid with a corner cell removed. This way, we can enforce strict strategies based on casework. For a $3 \times n$ grid, suppose we place a vertical tile at the top left corner; we then now have to put a horizontal tile under it, resulting in the configuration of a $s(n-1)$. Now, had we placed a horizontal tile, then we would have to place 2 vertical tiles under it, thus $t(n-2)$. With similar reasoning, we yield out

$$t(n) = t(n-2) + s(n-1)$$

$$s(n) = t(n-1) + s(n-2)$$

Solving yields

$$t(n) = 4t(n-2) - t(n-4)$$

and we can now recursively find $t(8)$ with $t(0) = 1, t(2) = 3$, so $t(8) = 153$.

HARD | Tags: Probability | Topics: Continuous Random Variables | Date: June 12, 2026

65. Spacious Uniform Values I

Problem

You sample 101 uniformly random numbers in the interval $(0, 1)$. Find the probability that no two of the values selected are within a distance of $\frac{1}{1000}$ of one another. The answer should be in the fully reduced form $\left(\frac{a}{b}\right)^c$. Find $a + b + c$.

Solution. Suppose we have points $x_1, x_2, ...x_{101}$ that are ordered properly and have a distance of at least $\frac{1}{1000}$ apart. The key to this problem is to deploy a transformation by having

$$y_i = x_i + (i-1)d$$

where d is the distance apart. Thus, we go from

$$0 \leqslant x_1 < x_2 < \cdots < x_{101} \leqslant 1$$

Where $x_{i+1} - x_i > d$. With this substitution, we simply have $y_{i+1} > y_i$ and

$$0 \leqslant y_1 < y_2 < \cdots < y_{101} \leqslant 1 - (N-1)d$$

where $N = 101$. Then, we can find the area under the curve of y as $(1-(N-1)d)^N \times N!$. Since there's $N!$ ways of arranging the y's, we have that our answer is simply

$$(9/10)^{101}$$

so the sum is 120.

HARD | Tags: Probability | Topics: Discrete and Continuous Random Variables | Date: June 13, 2026

66. Coin Flipping Competition II

Problem

Ty, Guy, and Psy are all flipping fair coins until they respectively obtain their first heads. Let T, G, and P represent the number of flips needed for Ty, Guy, and Psy, respectively. Find $P[T \leqslant G \leqslant P]$.

Solution. Note that our PMF is equal to

$$P(T = a, G = b, P = c) = \frac{1}{2^{a+b+c}}$$

Our solution is simply

$$\sum_{i=1}^{\infty}\sum_{j=1}^{i}\sum_{k=1}^{j} P(T=k, G=j, P=i) = 8/21$$

as desired.

HARD **Tags: Probability** **Topics: Conditional Probability, Discrete and Continuous Random Variable** **Date:**

67. June 14, 2026

Problem

Suppose that we have 2 cars parked in a line occupying spaces 1 and 2 of a parking lot. Spaces 3 and 4 are initially empty. Every minute, a car is considered eligible to move forward one space if a) the space in front of them is empty prior to any potential moves and b) the label of the space they are on is less than 4. Suppose that every minute, if a car is eligible to move, it moves forward one space with probability $\frac{1}{2}$. What is the expected number of minutes before the cars occupy spaces 3 to 4?

Note: The space in front must be empty before any moves. For example, on the first turn, only the car in space 2 is eligible to move.

Solution. Let E_{ij} denote the expected number of additional minutes to reach the target state $(3, 4)$ given the cars occupy spaces i and j.

Starting from $(1, 2)$, only the car in space 2 is eligible (space 3 is empty, its label < 4; the car in space 1 is blocked since space 2 is occupied). It moves with probability $\frac{1}{2}$, so

$$E_{12} = 1 + \tfrac{1}{2}E_{12} + \tfrac{1}{2}E_{13} \implies E_{12} = 2 + E_{13}.$$

From $(1, 3)$, both cars are eligible (space 2 empty in front of car 1, space 4 empty in front of car 3), and each moves independently with probability $\frac{1}{2}$:

$$E_{13} = 1 + \tfrac{1}{4}\left(E_{24} + E_{23} + E_{14} + E_{13}\right).$$

The remaining states resolve by the same one-step conditioning:

$$\begin{gathered} E_{24} = 1 + \tfrac{1}{2}E_{24} + \tfrac{1}{2}E_{34} \implies E_{24} = 2, \\ E_{14} = 1 + \tfrac{1}{2}E_{14} + \tfrac{1}{2}E_{24} \implies E_{14} = 4, \\ E_{23} = 1 + \tfrac{1}{2}E_{23} + \tfrac{1}{2}E_{24} \implies E_{23} = 4, \\ E_{34} = 0. \end{gathered}$$

Substituting into the equation for E_{13}:

$$E_{13} = 1 + \tfrac{1}{4}\left(2 + 4 + 4 + E_{13}\right) \implies \tfrac{3}{4}E_{13} = 1 + \tfrac{10}{4} \implies E_{13} = \tfrac{14}{3}.$$

Therefore

$$E_{12} = 2 + \frac{14}{3} = \boxed{\frac{20}{3}}.$$

HARD **Tags: Probability** **Topics: Combinatorics, Discrete Random Variables** **Date: June 15, 2026**

68. Doubly 5 I

Problem

Kathy has a fair 6-sided die with numbers 1-6 on the sides. She keeps rolling, keeping track of the outcomes, until she gets both a 4 and a 6. Find the probability Kathy observed exactly two 5s while rolling her die.

Solution. Ignore all rolls that are $1, 2, 3$ (they don't affect when we stop or the count of 5s). Among the "relevant" rolls $\{4, 5, 6\}$, each is equally likely, so condition on this reduced die where each of $4, 5, 6$ has probability $\frac{1}{3}$.

We roll (relevant rolls) until we have seen *both* a 4 and a 6, and we want exactly two 5s along the way.

Split by which of 4/6 appears first. By symmetry, assume WLOG a 4 appears before the first 6 (the 6-first case is symmetric and doubles the answer). The stopping roll is a 6. So the sequence of relevant rolls looks like:

$$\underbrace{\{4,5\}^*}_{\text{before first 4}} \quad \text{(with a 4 landing)} \quad \underbrace{\{4,5\}^*}_{\text{after 4, before 6}} \quad 6,$$

i.e. every relevant roll before the terminating 6 is a 4 or a 5, at least one is a 4, and exactly two are 5s.

More directly: consider the relevant rolls before the final 6. Each is 4 or 5 (never a 6, since the 6 terminates). We need exactly two 5s among them and at least one 4.

Let the terminating 6 be preceded by a string of 4s and 5s containing exactly two 5s and at least one 4. If there are $m \geqslant 1$ fours, the string has length $m + 2$, and the two 5s can be placed in $\binom{m+2}{2}$ ways among the first $m + 2$ positions; the last roll is the 6. Each specific string of length $m + 3$ has probability $\left(\frac{1}{3}\right)^{m+3}$.

Summing (and doubling for the symmetric 6-first case):

$$P = 2 \sum_{m=1}^{\infty} \binom{m+2}{2} \left(\frac{1}{3}\right)^{m+3}.$$

Evaluating the sum gives

$$P = \boxed{\frac{19}{108}}.$$

MEDIUM Tags: Pure Math Topics: Stochastic Calculus Date: June 16, 2026

69. Integral Variance II

Problem

Let W_t be a standard Brownian motion. Compute $\operatorname{Var}\left(\int_0^t W_s \, dW_s\right)$. The answer is in the form kt^2 for a constant k. Find k.

Solution. By Itô's formula applied to $f(W_t) = W_t^2$,

$$d(W_t^2) = 2W_t \, dW_t + dt \implies \int_0^t W_s \, dW_s = \frac{1}{2}W_t^2 - \frac{1}{2}t.$$

Thus

$$\mathrm{Var}\left(\int_0^t W_s\,dW_s\right) = \mathrm{Var}\left(\tfrac{1}{2}W_t^2 - \tfrac{1}{2}t\right) = \frac{1}{4}\mathrm{Var}(W_t^2).$$

Since $W_t \sim \mathcal{N}(0,t)$, we have $\mathbb{E}[W_t^2] = t$ and $\mathbb{E}[W_t^4] = 3t^2$, so

$$\mathrm{Var}(W_t^2) = \mathbb{E}[W_t^4] - \left(\mathbb{E}[W_t^2]\right)^2 = 3t^2 - t^2 = 2t^2.$$

Therefore

$$\mathrm{Var}\left(\int_0^t W_s\,dW_s\right) = \frac{1}{4}\cdot 2t^2 = \frac{t^2}{2}.$$

Hence $k = \boxed{\frac{1}{2}}$.

HARD Tags: Probability Topics: Combinatorics Date: June 17, 2026

70. Consecutive Pairs

Problem

Consider the set of 10 consecutive integers $\{1, 2, \ldots, 10\}$. How many subsets contain exactly one pair of consecutive integers? For example, $\{3, 5, 6, 9\}$ contains exactly one pair of consecutive integers.

Solution. Consider turning this problem into a butch of blocks. Suppose our subset consists of m numbers; then we have $m-2$ singleton blocks and 1 domino block. Now, to create this, we must use $m-1$ numbers to be placed between them. We then have $10-(m-1)-(m-1) = 12-2m$ numbers remaining to place in a total of m different spots. The number of ways to do this is equal to

$$\binom{12-2m+(m-1)}{m-1} = \binom{11-m}{m-1}$$

and since m can range from 2 to 5, we sum them up to get 235 as our answer.

HARD Tags: Probability Topics: Expected Value, Continuous Random Variable Date: June 18, 2026

71. Sum Exceedance I

Problem

Let $X_1, X_2, \ldots$ be IID Unif$(0,1)$ random variables and let $N = \min\{n : X_1 + \cdots + X_n > \ln(2)\}$. Find $\mathbb{E}[N]$.

Solution. Let $f(t)$ be the expected time for $X_1 + X_2 + \ldots + X_n > t$ where t is between 0 and 1, we then have

$$f(t) = 1 + \int_0^t f(t-x)\,dx = 1 + \int_0^t f(u)\,du$$

$$f(t) = e^t$$

And so our answer is 2.

HARD Tags: Probability Topics: Expected Value Date: June 19, 2026

72. Sum Exceedance II

Problem

Let $X_1, X_2, \cdots \sim \text{Unif}(0,1)$ IID and $N_2 = \min\{n : X_1 + \cdots + X_n > 2\}$. Find $\mathbb{E}[N_2]$. Your answer will be in the form $ae^2 + be$ for integers a and b. (e here is Euler's constant.) Find $a + b$.

Solution. This problem is similar to the past one, but that one only works if your t is between 0 and 1; here, it's greater than 1, so we just write our integral a bit differently. We have

$$f(t) = 1 + \int_0^1 f(t-x)\ dx = 1 + \int_{t-1}^t f(u)\ du$$
$$f'(t) = f(t) - f(t-1)$$

Note that $f(t-1)$ is between 0 and 1 since we desire $f(2)$, so

$$f'(t) - f(t) = e^{t-1} \to (e^{-t} f(t))' = e^{-1} \to f(t) = e^{t-1}t + ce^t$$

and since $f(1) = e$, we have

$$f(t) = e^t - e^{t-1}(t-1)$$

and so our answer is $f(2) = e^2 - e$ and so $a = b = 0$.

HARD Tags: Probability Topics: Games, Conditional Probability, Continuous Random Variables Date: June 20, 2026

73. Competitive Sampling

Problem

You and your opponent each draw a number from $U(0,1)$ without revealing it. You each have the option to redraw and keep that value instead, and the other person doesn't know which choice they made. The person with the higher number wins. The optimal strategy is of the form: reroll if the number is less than k. Find k to 3 decimal places.

Solution. At the breakeven point k, standing on the value k and rerolling give the same winning probability — that is what makes k optimal. We check both sides against an opponent using the same threshold k, whose final value Y has CDF

$$F(x) = \begin{cases} kx, & x < k \\ kx + (x-k), & x \geqslant k. \end{cases}$$

Standing on k: you win when $Y < k$, so $P(\text{win}) = F(k) = k^2$. This is the left side.

Rerolling: you get a fresh $Z \sim U(0,1)$ and win with probability $F(Z)$, averaged over Z:

$$\int_0^1 F(z)\,dz = \underbrace{\int_0^k kz\,dz}_{k^2/2 \text{ (rescaled)}} + \underbrace{\int_k^1 kz\,dz}_{(k+1)(1-k^2)/2 \text{ part}} + \underbrace{\int_k^1 (z-k)\,dz}_{-k(1-k) \text{ correction}} .$$

Setting the two equal gives the equation for k:

$$k^2 = \frac{k^2}{2} + \frac{(k+1)(1-k^2)}{2} - k(1-k).$$

The right side is just $\int_0^1 F(z)\,dz$ regrouped: the $\frac{k^2}{2}$ comes from the reroll region below k, the $\frac{(k+1)(1-k^2)}{2}$ bundles the kz and z pieces above k, and $-k(1-k)$ subtracts the $-k$ offset in the $(z-k)$ term over $[k,1]$. Simplifying yields

$$k^2 + k - 1 = 0 \implies k = \frac{\sqrt{5}-1}{2} \approx 0.618.$$

HARD Tags: Probability Topics: Expected Value, Games Date: June 21, 2026

74. Unknown Starter

Problem

You have 3 cards, each labeled n, $n+1$, $n+2$, and you don't know n. All cards start face down. You flip one card and observe the value. If you say "stay", your payout is equal to that card's value. If you don't "stay", then you flip another card. Again, choose to "stay" (and receive payout equal to the second card's value) or flip the final card and receive payout equal to the final card's value. Design the optimal strategy. The expected payout of this strategy is $n+c$ for some constant c. Find c.

Solution. The key idea is to consider the scenarios in 3 cases. To have an estimate idea of what n is, we will always need to flip twice. So consider these cases: $n+2, n$ or $n, n+2$: The 2 gap allows us to infer what the next number will be, so we always flip in the former scenario and don't flip for the latter. This yields $1/6(1+2)$ contribution to our expected value
$n, n+1$ or $n+1, n+2$. In this scenario, from our perspective, we basically got $a, a+1$, and the flip would either yield $a-1$ or $a+2$ each with probability $1/2$. If we roll, our expected value is $a+1/2$, which is less than $a+1$, so we never reroll for this case. Thus, this contributes $1/6(1+2)$ to our expected value.
Finally, if we get $n+1, n$ or $n+2, n+1$, from our perspective, we got $a, a-1$, and the flip would either yield $a+1$ or $a-2$, and the expected value of this $a-1/2$, which is greater than $a-1$, so we always flip. Thus, this contributes $1/6(2)$ to our expected value. Adding those contributions yield $4/3$

HARD Tags: Brainteasers Topics: None Date: June 21, 2026

75. Water Measurement

Problem

A maid was sent to the brook with two vessels that measured exactly 7 pints and 11 pints. She had to bring back exactly 2 pints of water. What is the smallest possible number of transactions necessary? A transaction is filling a vessel, emptying it, or pouring from one vessel to another. In other words, you cannot transfer a proportion of the vessel — you must transfer all of it.

Solution. 14 transactions.

One optimal sequence (writing states as (7-pint, 11-pint), starting from $(0, 0)$):

1. Fill 7: $(7, 0)$
2. Pour $7 \to 11$: $(0, 7)$
3. Fill 7: $(7, 7)$
4. Pour $7 \to 11$: $(3, 11)$
5. Empty 11: $(3, 0)$
6. Pour $7 \to 11$: $(0, 3)$
7. Fill 7: $(7, 3)$
8. Pour $7 \to 11$: $(0, 10)$
9. Fill 7: $(7, 10)$
10. Pour $7 \to 11$: $(6, 11)$
11. Empty 11: $(6, 0)$
12. Pour $7 \to 11$: $(0, 6)$
13. Fill 7: $(7, 6)$
14. Pour $7 \to 11$: $(2, 11)$

The 2 pints remain in the 7-pint vessel after 14 transactions, which a breadth-first search over all reachable states confirms is the minimum.

HARD | Tags: Probability | Topics: Continuous Random Variables | Date: June 22, 2026

76. Forming a Triangle

Problem

A stick of 1 metre is randomly broken into three pieces. Given each break follows a uniform distribution along the stick, what is the probability that the three segments can form a valid triangle?

Solution. We deploy our geometric probability skillset to this problem. Consider x being our first break and y being our second break with $y > x$. We then have lengths $x, y - x, 1 - y$, putting them on the constraint of triangle inequality yields

$$y > 1/2$$

$$x < 1/2$$

$$y < x + 1/2$$

Plotting this in the graph and finding the area of this divided by that of the total region (which is the area bounded by $x = 1, y = 1, y > x$) yields $1/4$ as our answer.

HARD Tags: Probability Topics: Games Date: June 23, 2026

77. 2 Below I

Problem

You and a friend play a game where you both select an integer 1-100. The winner receives \$1 from the loser. The winner is the player who selects the strictly higher number. If there is a tie, then nothing happens. However, a player can also win by selecting a value exactly 2 below the larger integer. For example, if you select 80 and your friend selects 82, you are the winner in this case. Assume both you and your friend play optimally. The optimal strategy here is a mixed strategy, where you select a random value X from some appropriately determined distribution. Find $\text{Var}(X)$.

Solution. You would only want to choose a number from $98, 99, 100$ since if you choose n, $n+3$ would always be a better choice. Thus, choose one of the 3 with equal probability, and so our answer is $2/3$.

HARD Tags: Probability Topics: Expected Value, Games Date: June 24, 2026

78. Optimal Marbles I

Problem

Two players, A and B, play the following game: Both players have 100 marbles and may put anywhere between 1 and 100 marbles in the box each. This decision is not revealed to the other player. Then, they draw 2 marbles with replacement between trials. If a drawn marble belongs to A, then, assuming A put a marbles in the box, A is paid $100 - a$ monetary units from a third party. Similarly, if a drawn marble belongs to B, then, assuming B put b marbles in the box, B is paid $100 - b$ monetary units from a third party. Assume both players play optimally. Find the expected total payout of player A.

Solution. Let's write out the expected payout for 1 marble and multiply by 2 at the end. We have $A(a, b) = a/(a + b)(100 - a)$ and differentiaying and setting to 0 yields

$$100b - 2ab - a^2 = 0$$

The reason we set this to 0 is the find the local maximum of this equation. Note that we can now set $a = b$ because at equilibrium, both users would choose the same thing, we get

$$a = 33.333$$

so we test $a = b = 33$ to yield the our answer as 67

HARD Tags: Brainteasers Topics: None Date: June 25, 2026

79. Clockwise Murder

Problem

There are N people standing in a circle labeled 1 through N, such that the smallest power of two less than or equal to N is X. Starting with person 1, they take a sword and kill the person to their left. Then, they pass the sword to the living person on their left, and the process continues until one person is left. The survivor's number can be expressed as $aN + bX + c$. Compute $a + 2b + 3c$.

Solution. The key to this problem is strong induction and applying the fact that we can simply shift numbers if needed. If N is even, we get

$$1, 3, 5, 7, 9, \ldots N - 1$$

after the first pass, and if N is odd, we get

$$3, 5, 7, 9, \ldots N$$

If $J(N)$ is the position of the survivor, then we have

$$J(N) = 2J(N/2) - 1$$

if N is even, and

$$J(N) = 2J((N-1)/2) - 1$$

if N is odd. Now, let $N = X + L$. After some induction, we get that $J(N) = 2L + 1 = 2N - 2X + 1$ and so our answer is 1

HARD Tags: Probability Topics: Combinatorics Date: June 26, 2026

80. 1 Glove Off

Problem

You have 5 pairs of gloves that each have a distinct number $1-5$. The 10 gloves are randomly paired up. Find the probability that the gloves are paired up such that the values of any pair differ by at most 1.

Solution. Note for choosing pairs, we can choose $(2n)!/(2^n \cdot n!)$ where $n = 5$. Let g_n be the number of valid pairs. If n is paired with n, this gives g_{n-1} contribution, and if n is paired with $n-1$, we have 2 ways to do this, and it yields $2g_{n-2}$, so we have

$$g_n = g_{n-1} + 2g_{n-2}$$

and $g_1 = g_0 = 1$, solving for $g_5 = 21$, so answer is $21/945 = 1/45$. Note that we have two ways because the two gloves for each pair are distinguishable (right and left hand glove).

HARD | Tags: Probability | Topics: Continuous Random Variable, Expected Value, Games | Date: June 27, 2026

81. RNG on RNG

Problem

You generate a uniformly random number in the interval $(0, 1)$. You can generate additional random numbers as many times as you want for a fee of \$0.02 per generation. This decision can be made with the information of all of the previous values that have been generated. Your payout is the maximum of all the numbers you generate. Under the optimal strategy, find your expected payout of this game.

Solution. To solve this problem, we consider the threshold T that we would stop at. Suppose our current max is x, then the expected new max would be

$$(1 - x)((x + 1)/2) + x \times x = (1 + x^2)/2$$

and we need this change in improvement to be greater than the change in cost, 0.02, so we have

$$(1 + x^2)/2 - x > 0.02$$

and this yields $x > 0.8$, so terminate at $T = 0.8$. Now, we solve this by considering what happens at our first toss. Let's consider our expected return V with $T = 0.8$, we then have

$$V = -0.02 + TV + (1 - T)(1 + T)/2$$

and get $V = 0.8$, but this is for games that are onward after the first game that doesn't cost anything, so our answer is

$$(0.2)(0.9) + (0.8)(0.8) = 0.82$$

where the former is if we get greater than 0.8 first try and latter is if we don't.

MEDIUM | Tags: Probability | Topics: Combinatorics | Date: June 28, 2026

82. All Attainable Values

Problem

How many 6-sided dice with values on each side in the set $\{1, 2, 3, 4, 5, 6\}$ are there with the property that when rolled twice, for each integer $2 \leqslant k \leqslant 12$, there is positive probability that the sum is exactly k? Note that not every value in the set necessarily needs to be used and that two dice are considered indistinguishable if they contain the exactly same amount of faces corresponding to each value in the set, regardless of the labelling of the sides. Assume each side appears with equal probability.

Solution. Note that to attain $2, 3, 11, 12$, we must have $1, 2, 5, 6$. Now, we need either 3 or 4. We can either have both or have only 1 of them. If we have only one of them, we have two choices for which one, and then 5 choices for who the extra is. For both, we have 1 way, so our answer is 11.

MEDIUM Tags: Conditional Expectations, Games Topics: Probability Date: June 29, 2026

83. Busted 6 II

Problem

Suppose you play a game where you continually roll a die until you obtain either a 5 or a 6. If you receive a 5, then you cash out the sum of all of your previous rolls (excluding the 5). If you receive a 6, then you receive no payout. You also have the decision to cash out mid-game for the sum of all your previously obtained rolls. Assuming optimal play, what is your expected payout? Round your answer to the nearest hundredth.

Solution. First, we find which point do we stop. We find this via

$$T \geqslant 1/6(T+1) + 1/6(T+2) + 1/6(T+3) + 1/6(T+4) + 0$$

and get $T \geqslant 10$, so we cash out if our total is greater than or equal to 10. Now, let $V(s)$ be the expected value if you start at s total. We have $V(s) = s$ for $s \geqslant 10$ and

$$V(s) = \frac{1}{6}(V(s+1) + V(s+2) + V(s+3) + V(s+4) + s)$$

and starting from 9 and going down, we get $V(0) = 3.03$, which is our answer.

HARD Tags: Brainteasers Topics: None Date: June 30, 2026

84. Stack Double

Problem

7 men labeled $A - G$ play a coin flip game that starts with player A. At each turn, the player will flip a coin. If it appears heads, then the player must double the stack of each other player that plays the game, taking the money from their own stack. In other words, the payout to each of the other 6 players is the value of their current stack. Otherwise, nothing happens. The sequence $HHHHHHHHHHHHHH$ is observed and each of the 7 players ends up with \$1.28 as their final stack. If $x_1, \ldots, x_7$ represent the amount that each player started with, find the value of $10000(x_1^2 + \cdots + x_7^2)$. For example, if player A starts with \$5.31, then $x_1 = 5.31$.

Solution. We solve this question by thinking backward. For the x_7, before giving away its portion, it has $64x_7$ coins, and the rest of the coins have $1.28/2$ since doubling each would lead to the final amount being 1.28. Thus, we have

$$64x_7 - 6(1.28/2) = 1.28 \to x_7 = 0.08$$

Similarily, for x_6, before giving away, it has $32x_6$, and has to give away $32x_7$ and also $5(1.28/4)$ with a similar logic, so we have

$$32x_6 - 32x_7 - 5(1.28/4) = 1.28/2$$

Continuing this yields $x_7 = 0.08, x_6 = 0.15, x_5 = 0.29, x_4 = 0.57, x_3 = 1.13, x_2 = 2.25, x_1 = 4.49$, so our answer is 269374

HARD Tags: Probability Topics: Conditional Probability Date: July 1, 2026

85. Voter Mayhem II

Problem

Two candidates, say A and B, are running for office. Candidate A received n votes, while Candidate B received m votes, with $n > m$. The $n + m$ votes are thrown into a box and shuffled around. Then, the votes are drawn without replacement one-by-one. A running tally of the number of votes for each candidate is kept. The probability that Candidate A is never behind in the voting count (excluding the initial state where both have 0) is a function $Q(n, m)$. Find $Q(100, 80)$.

Solution. We can use the reflection principle to solve this problem. Suppose getting an A vote counts as a $+1$ and B as a -1. Then our final destination is $n - m$, and we are only behind iff the graph hits -1 at some point. If we do, we can reflect the rest of the graph, which would lead to our final destination being $m - n - 2$. This is equivalent to how many ways we can choose $n + 1$ down from $n + m$ while the number of ways are choosing n ups from $n + m$ ways. Thus, our answer is the complement of this,

$$1 - \frac{\binom{n+m}{n+1}}{\binom{n+m}{n}} = 1 - \frac{m}{n+1}$$

so our answer is 21/101

HARD Tags: Probability Topics: Expected Value Date: July 2, 2026

86. Coloring Components III

Problem

Consider a line of 20 adjacent colorless squares. Color in each individual square black or white with equal probability, independent of all other squares. A connected 5-component is a group of 5 consecutive black squares. Note that overlapping components are not counted. For example, WBBBBBWWB and WBBBBBBWB have 1 connected 5-component, but WBBBBBBBBBBW has 2 connected 5-components. Find the expected number of connected 5-components in our line. The answer can be written as a simplified fraction of the form $\frac{p}{q}$. Find $p + q$.

Solution. Note that we can rewrite our expectation as

$$\sum_{i}^{4} P(N \geqslant i)$$

where N is the number of connected components. Suppose we have m components, we can compute the probability by trying each index as a starting point. For index 1, the probabiltiy is $1/2^m$ while for index greater than 1, the previous block must be white, so the probability is $1/2^{m+1}$, summing up a

$$1/2^m + (20 - m)1/2^{m+1}$$

noting that 20-m is neccessary from the last blocks not being able to be a start. Then, let $m = 5, 10, 15, 20$ and sum them up, and get $p = 284785, q = 1048576\text{i}$

HARD | Tags: Probability | Topics: Conditional Expectation, Discrete Random Variables | Date: July 3, 2026

87. Sequence Terminator

Problem

A fair 6-sided die is rolled repetitively, forming a sequence of values, under the following rules: If any even value is rolled, add it to the current sequence. If a 3 or 5 is rolled, discard the entire sequence and don't add the 3 or 5 to the start of the new sequence. If a 1 is rolled, add the 1 to the current sequence and end the game. Find the expected length of the sequence at the end of the game.

For example if the sequence of rolls is 34265241, we would reset to an empty sequence at the first roll, reset to an empty sequence at the 5, and our final sequence would be 241, which is of length 3.

Solution. Let $f(n)$ be the length of the sequence given we currently have n even rolls. We want $f(0)$. We have

$$f(n) = \frac{1}{6}(n+1) + \frac{1}{2}f(n+1) + \frac{1}{3}f(0)$$

where the first term is if we roll a 1, second term is if we roll an even number, and third is if we roll 3 or 5. Suppose $f(n) = an + c$, and we get $a = 1/3$, and $c = 1/2$ which is our answer

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MEDIUM | Tags: Probability | Topics: Combinatorics, Conditional Probability, Discrete Random Variable | Date: July 4, 2026

88. Exact 5 I

Problem

James continually rolls a fair 6-sided die until he obtains his first 6. Compute the probability that James obtains the value 5 exactly four times before he stops rolling.

Solution. Note that since we don't care about the other rolls, we can think of this problem simply as flipping a coin that has either 5 or 6, so the answer is $\frac{1}{32}$.

MEDIUM Tags: Probability Topics: Combinatorics Date: July 5, 2026

89. Power Digits

Problem

m is randomly selected from $\{111, 133, 155, 177, 199\}$. n is randomly selected from $\{2004, 2005, \ldots, 2023\}$. What is the probability that m^n has a 1 as the units digit?

Solution. If $m = 111$, we're good 1/5. If $m = 133$, we just need a multiple of 4 with probability 5/20, same for 177, and for 199, we need a multiple of 2, so 1/2. Thus, our answer is

$$1/5 + 1/4 \times 1/5 + 1/4 \times 1/5 + 1/2 \times 1/5 = 2/5$$

HARD Tags: Probability Topics: Expected Value Date: July 6, 2026

90. Car Crash

Problem

N employees are driving their individual cars on their way to work at QuantEssential. All cars are reasonably spaced apart, and they all travel at distinct speeds which are randomly assigned. When a faster car catches up to a slower car, it assumes the slower car's speed. After a very long period of time has passed, all N cars have settled into K clusters traveling at distinct speeds. What is the expected value of K when $N = 10$?

Solution. Consider from the most ahead car to the least ahead car. For the first one, it will always form a cluster, so it contributes 1, for the second most ahead, it will contribute 1/2 since its probability of being slower than the first car is 1/2, logic applies to the third with 1/3, and so on to 1/10. These are the indicator values, so the answer is just

$$1 + 1/2 + 1/3 + \cdots + 1/10$$

MEDIUM Tags: Probability Topics: Expected Value Date: July 7, 2026

91. Repetitious Game II

Problem

Audrey repeatedly draws cards from a standard 52-card deck with replacement. Compute the expected number of draws for Audrey to get at least one card from each suit.

Solution. Note that since this has replacement, we can think of the probability of getting a specific suit as 1/4. Thus, for the first card, we have the expected value of 1 to get a card from a different suit. Now, for the second suit, we simply consider the probability as 3/4, and since this is a geometric probability, the expected value for the second suit is 4/3, and the 4/2 for the third, and 4/1 for the last. Thus, our answer is $4(1 + 1/2 + 1/3 + 1/4) = 25/3$

HARD | Tags: Probability | Topics: Expected Value, Conditional Expectation | Date: July 8, 2026

92. Sum Exceedance IV

Problem

A fair 5-sided die with the values 1-5 on the sides is rolled repeatedly until the sum of all upfaces is at least 5. Find the expected number of times the die needs to be rolled.

Solution. This is just a Markov chain. We have

$$E(0) = 1 + 1/5(E(1) + E(2) + E(3) + E(4) + E(5)$$
$$E(1) = 1 + 1/5(E(2) + E(3) + E(4) + E(5))$$
$$E(2) = 1 + 1/5(E(3) + E(4) + E(5))$$
$$E(3) = 1 + 1/5(E(4) + E(5))$$
$$E(4) = 1, E(5) = 0$$

And solving this upward yields $E(0) = 1296/625$. Note we could've done this even nicer with recursion and some symmetry to get a closed form (note how it's $(6/5)^4$)

HARD | Tags: Probability | Topics: Games, Conditional Probability | Date: July 9, 2026

93. Free Sundae

Problem

You are in line at an ice cream shop when the manager announces that she will give a free sundae to the first person in line whose birthday is the same as someone who has already bought an ice cream. Assuming that you do not know anyone else's birthday and that all birthdays are uniformly distributed across the 365 days in a normal year, what position in line will you choose to maximize your probability of receiving the free sundae?

Solution. We have that the probability of receiving free ice cream if we line at k is $P(k) = (365!(k-1)/(365^k(366-k)!)$. For the value to be maximized, we have $P(k) > P(k-1)$ and $P(k+1) < P(k)$. Solving this, we get $19.61 < k < 20.61$, so our answer is $k = 20$

HARD | Tags: Probability | Topics: Expected Value, Games | Date: July 10, 2026

94. Take And Roll II

Problem

You are playing a game with a total of 100 actions and a fair 20-sided die.

The die starts with face of 1. The two actions you can perform in the game are to roll and to take. Performing a roll re-rolls the current face of the die. Performing a take allows you to cash out the current face of the die. The game does ends when you complete all 100 actions and you must roll the die again before doing another take. Your strategy is to accept any number that is at least some threshold n. This n must be decided in advance and is fixed for the entire game. Assuming rational play in selecting n, find your expected payout.

Solution. Suppose the probability of getting a number at least the threshold is $p = (21 - n)/20$, then the expected number of actions required would be $1/p + 1$, and so we would get to cash out a total of

$$\frac{100}{\frac{1}{p} + 1} = \frac{100p}{1 + p}$$

times. Each time, we can receive an expected amount of $(n + 20)/2$, so multiply the expression by this. Now simply try values of n to see that $n = 6$ yields the most amount of money with 557 dollars.

HARD Tags: Probability Topics: Combinatorics Date: July 11, 2026

95. Game Time

Problem

38 people waiting to see a tennis match. The entrance fee is \$5 to enter. 19 of the people only have \$10 bills, while the other 19 only have \$5 bills. The cashier currently has no change. If the 38 people arrange themselves in line completely at random, what is the probability that all of the people can successfully purchase a ticket without reordering in line.

Solution. This is another typically random walk problem. Consider up a 5 dollar bill and down a 10 bill. We model it as how we can reach 0 after 38 of such steps. We need to reorder if our function ever touches -1, which is the same as the number of paths of reaching 38 if you start -2, which has $\binom{38}{20}$ ways while the total is $\binom{38}{19}$, so our answer is $1 - 19/20 = 1/20$

HARD Tags: Brainteasers Topics: None Date: July 12, 2026

96. Cats and Mice

Problem

A number of cats (more than one) killed between 999919 mice, and every cat killed an equal number of mice (more than one). Each cat killed more mice than there were cats. How many cats were there?

Solution. Let there be c cats that each killed k mouses. We then have

$$ck = 999919 = 1000^2 - 9^2 = 1009 \times 991$$

and since $c < k$, we have $c = 991$

HARD Tags: Brainteasers Topics: None Date: July 13, 2026

97. Consecutive Children

Problem

9 children of distinct ages (in years) were born with a fixed interval of time between consecutive children. The sum of the squares of the ages of all the children is equal to the square of their father's age. Assuming that the father is aged at most 60, what is the father's age?

Solution. Let the ages of the children be $m - 4s, m - 3s, \ldots m, \ldots, m + 4s$ and the age of the father be F. We then have

$$9m^2 + 60s^2 = F^2$$

SInce m, s, F are all integers, F must be divisible by 3, and then it follows that s must be divisible by 3 as well, so let $F = 3a, s = 3b$ and yield

$$m^2 + 60b^2 = a^2 \rightarrow (a - m)(a + m) = 60b^2$$

We now do casework. If $b = 1$, we have that $a = 16$ or $a = 8$. The former works while the latter results in a negative age for the last children. For other $b > 1$, we get that we will get negative ages, so our answer is $F = 3a = 48$

HARD Tags: Probability Topics: Games, Expected Value Date: July 14, 2026

98. Place or Take

Problem

You are playing a one-player game with two opaque boxes. At each turn, you can choose to either "place" or "take". "Place" places \$1 from a third party into one box randomly. "Take" empties out one box randomly and that money is yours. This game consists of 100 turns where you must either place or take. Assuming optimal play, what is the expected payoff of this game? Note that you do not know how much money you have taken until the end of the game.

Solution. No place should follow a take: any place after a take only risks landing in an already-emptied box, while every place before a take can add to the pool. Hence all places come first, then all takes. Let p be the number of places and $t = 100 - p$ the number of takes.

After p places, each place adds one dollar to a uniformly random box, so if box A holds k, box B holds $p - k$ with $k \sim \text{Binomial}(p, 1/2)$. The two boxes total p.

For the t takes, each take empties a random box (resetting it to 0); with no refills, a box hit more than once yields zero after the first. The total collected is the sum of the boxes hit at least once. Both boxes hit has probability $1 - 2^{1-t}$, yielding the full p; only one box hit has probability 2^{1-t}, yielding an expected $p/2$ by symmetry.

The expected payoff is

$$E(p) = (1 - 2^{1-t})\,p + 2^{1-t} \cdot \frac{p}{2} = p\left(1 - 2^{-t}\right) = p\left(1 - 2^{-(100-p)}\right).$$

Evaluating near the maximum: $(96, 4)$ gives 90.00, $(95, 5)$ gives 92.03, $(94, 6)$ gives 92.53, and $(93, 7)$ gives 92.27. The optimum is $t = 6$, $p = 94$:

$$E = 94(1 - 2^{-6}) = 94 \cdot \frac{63}{64} \approx 92.53.$$

HARD Tags: Probability Topics: Expected Value, Games Date: July 15, 2026

99. Optimal Marbles

Problem

Two players, called X and Y, engage in a strategic game:

Each has 100 tokens and secretly decides how many tokens, between 1 and 100, to contribute to a shared pool. Once both players have made their decisions, one token is randomly drawn from the pool. If the token belongs to X, then assuming X contributed x tokens, X receives $100 - x$ monetary units from Y.

Similarly, if the token belongs to Y, then assuming Y contributed y tokens, Y receives $100 - y$ monetary units from X. Both players aim to maximize their respective monetary outcomes and are assumed to play optimally. How many tokens should X choose to contribute?

Solution. Suppose X contributes x and Y contributes y, the expected final value X gains is then

$$\frac{x}{x+y}(100+100-x) - \frac{y}{x+y}(100-(100-y)) = 200\frac{x}{x+y} + (y-x) = 200(1-\frac{y}{x+y}) + y - x$$

Taking the derivative with respect to x to get the max value, we have

$$200\frac{y}{(x+y)^2} - 1 = 0$$

Now, since both play optimally, at this step, we can assume that $x = y$. We have

$$200/4x - 1 = 50/x - 1 = 0 \rightarrow x = 50$$

HARD Tags: Probability Topics: Expected Value Date: July 16, 2026

100. Marble Runs

Problem

You repeatedly draw marbles from a bag containing 50 red and 50 blue marbles until there are no more marbles left, recording the order of red and blue marbles drawn. You then count the number of "runs," where a run is defined as any number of consecutive marbles of the same color. For example, $RBBRRRBRR$ contains 5 runs. What is the expected number of runs that you observe?

Solution. Consider the start of each run. The first marble always starts a new run and thus contributes 1 to the overall number of runs. Consider X_i now for $i > 1$ where X_i is the indication function that marble i and $i-1$ are difference, meaning i contributes to a new run. This occurs with probability $2 \times 50/100 \times 50/99 = 50/99$ from placing the two marbles in the line of 100 at the beginning and considere RB and BR. By linearity of expectation, we then have

$$1 + 99 \times 50/99 = 51$$

HARD | Tags: Probability | Topics: Conditional Probability, Discrete Random Variable | Date: July 17, 2026

101. Finite Coin Equalizer

Problem

Suppose you flip a coin that has probability $0 < p < 1$ of landing heads on each flip. You flip the coin until you have an equal amount of heads and tails appearing for the first time (of course, after the initial state of 0 for both). For any integer $n \geqslant 1$, find the probability that you flip the coin exactly $2n$ times. The probability can be written as a function $f(p, n)$. Compute $f(1/3, 5)$ to the nearest ten-thousandth.

Solution. The key is to notice that the problem is basically a random walk. Start with the fact that each path that is length $2n$ has probability $(p(1-p))^n$. Next, we find the number of paths that bring us there. WLOG, suppose the first step is heads. We multiply this number of paths by 2 at the end since it can also be tails. Then, note that we need to SHIFT where we do the reflection because BOTH endpoints are on the reflection line $y = 0$ and that's not good. Thus, note that the last coin must be a tails if the first is heads. We then have $2n-2$ coins left to decide the sequence of. Note that now, we basically start at $y = 1$ (assuming heads is $+1$), and then we must go to 1 again in $2n-2$ steps without ever crossing $y = 0$. This is a lightwork reflection problem now, and we have

$$f(p, n) = 2\left[\binom{2n-2}{n-1} - \binom{2n-2}{n-2}\right]\Big(p(1-p)\Big)^n$$

HARD | Tags: Probability | Topics: Combinatorics, Conditional Probability | Date: July 18, 2026

102. Empty Urn

Problem

We flip a fair coin until we obtain our first heads. If the first heads occurs on the kth flip, we are given k balls. We put them into 3 bins labeled 1, 2, and 3 at random. Find the probability that none of the three bins are empty.

Solution. Condition on k balls and multiple by $1/2^k$ at the end. First, by PIE, the probability that at least one bin is empty is

$$(3 \times 2^k - 3 + 0)/3^k$$

And so the probability is 1 minus that. Then we sum over all possible values of k, which is 1 to ∞, and get $1/10$ as our answer.

HARD Tags: Probability Topics: Conditional Probability Date: July 19, 2026

103. Limited Urns

Problem

Assume you have n identical urns, each containing red and blue balls. The ith urn, where $1 \leqslant i \leqslant n$, contains 1 red ball and $2^i - 1$ blue balls.

Suppose you randomly select an urn and select a ball from it, and its red. Let $p(n)$ be the probability that if you replace this ball and again draw a ball at random from the same urn then the ball drawn on the second occasion is also red. Compute $\lim_{n\to\infty} p(n)$.

Solution. Suppose we chose urn i with probability $1/n$, then the probability of this occurring is $1/4^i$. We sum this from 1 to n so we have

$$p(n) = \frac{1/n \sum_{i=1}^{n} 1/4^i}{1/n \sum_{i=1}^{n} 1/2^i} = 1/3(1 - 4^{-n})/(1 - 2^{-n})$$

and as $n \to \infty$, we have $1/3$ as our answer

MEDIUM Tags: Probability Topics: Continuous Random Variable Date: July 20, 2026

104. Sine Condition

Problem

$X \sim \text{Unif}([0, \pi])$. Find $\cos\big(E[X \mid \sin(X)]\big)$.

Solution. Recall that

$$\mathbb{E}(X|\sin(X)) = \sum_s \mathbb{E}(X|\sin(X) = s)P(\sin(X) = s)$$

Now, if $\sin(X) = s$, then our two possible values satisfy $x_1 + x_2 = \pi$ because of arcsin. Then, the expected value is equal to $\pi/2$. Now if sum them up for all s, then the overall expected value is just $\pi/2$, so the cos of that is just 0.

HARD Tags: Probability Topics: Expected Value, Continuous Random Variable Date: July 21, 2026

105. Segment Traversal

Problem

You select a uniformly random starting point on the circumference of a circle, say P_1. Then, you select another uniformly random point on the circumference, P_2, and draw a line segment between P_1 and P_2. You then continue to select uniformly random points on the circumference of the circle, say $P_3, \dots, P_n$, and draw a line segment between P_i and P_{i+1} for each $1 \leqslant i \leqslant n-1$. Lastly, you draw a line segment from P_n back to P_1. Find the expected number of intersections between line segments on the interior of the circle as a function of n. Evaluate this for $n = 12$.

Solution. This problem requires the use of linearity of expectation. We condition on every non-adjacent pair of segments, and there are $\binom{n}{2} - n = n(n-3)/2$ such pairs. For each pair, there is a probability of 1/3 that they intersect (think abiout placing 2 chords with 4 points. Only one of the three ways results in a intersection). Thus, our answer is $n(n-3)/6$